



Faculty of Computing and Information Technology (FCIT)

Machine Learning

ASSIGNMENT # 1

BS(SE)-Fall 2023

Instructor: Dr. Madiha Khalid

Total Marks: 100

Teacher Assistant: Rimsha Majeed

Deadline: Feb 17, 2026 12:15pm

Linear Regression

Instruction:

- Work must be your own. No copy, no plagiarism (0 marks if found).
- Use libraries like numpy, pandas, matplotlib, seaborn only for data handling and plotting.
- Do not use built-in regression models (e.g., `sklearn.linear_model.LinearRegression`).
- Submit as a Colab/Jupyter Notebook with proper comments.

Objective:

The purpose of this assignment is to help you understand the working of **Linear Regression** by implementing it from scratch in Python. You will load datasets, define the hypothesis function, implement the cost function, apply gradient descent to optimize parameters, and visualize the results.

To Do:

1. Dataset Load and Preprocess
2. Implement Linear Regression
3. Implement Gradient Descent
4. Visualization

Linear Regression

A supervised machine learning algorithm that models the relationship between a dependent (target) variable and one or more independent (predictor) variables by fitting a straight line (regression line).

Univariate Dataset (one feature, one target)

You are given a **real estate dataset**.

Real estate is one of those examples that every regression course goes through as it is extremely easy to understand and there is a (almost always) certain causal relationship to be found.

The data is located in the file: `'real_estate_price_size.csv'`.

You are expected to create a simple linear regression using the data.

In this , the dependent variable is 'price', while the independent variables is 'size'.

Part 1: Dataset Load and Preprocessing

(Marks: 30 = 10 + 20)

Use the **dataset files** (CSV, Excel, etc.) downloaded and saved locally.

Dataset: `(real_estate_price_size.csv)`

Preprocessing:

Before applying linear regression, you must preprocess the dataset:

- ✓ Handle Missing Values
- ✓ Feature Scaling (Normalization/Standardization)
- ✓ Splitting Data (i.e: Train, Test)

Part 2: Implementing Linear Regression

(Marks: 40 = 20 + 20)

You need to implement the following functions in Python.

1. Hypothesis Function

For Simple Linear Regression:

- → predicted value (what the model guesses).
- → input value.

- → slope (weight/parameter, tells how steep the line is).
- → intercept (bias/parameter, tells where the line cuts the y-axis).

2. Cost Function

Use **Mean Squared Error (MSE)** as the cost function:

- → cost (total error we want to minimize).
- → number of data points.
- → actual value.
- → predicted value for input.
- The square ensures errors are always positive.

Part 3: Optimization Function (Gradient Descent)

(Marks: 20)

Use **Gradient Descent** to minimize the cost function:

- → “update” (means we replace old value with new)
- → learning rate (small step size).
- → how much cost changes when slope m changes
- → how much cost changes when intercept c changes.

Compute Optimal Parameters

- Run your gradient descent implementation.
- Report the final θ values (parameters) and minimized cost.

Part 3: Visualization

(Marks= 10)

Visualize the Actual vs Predicted result

Note: Explore `matplotlib.pyplot` by yourself to know how to visualize the results

Helping Content: <https://www.geeksforgeeks.org/data-visualization/data-visualization-using-matplotlib/>

i.e: The following Sample code snippet is to compare **actual vs predicted values**:

Sample code:

```
import matplotlib.pyplot as plt
def plot_regression_line(X, y, y_pred):
    plt.scatter(X, y, color="blue", label="Actual
    Prices")

    plt.plot(X, y_pred, color="red",
    label="Predicted Prices")
```

